

Toffoli resource estimate: BSE effective Hamiltonian (diamond, $6 \times 6 \times 6$) — raw vs. symmetry-adapted ISDF–THC

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Setup and method

Multi-exciton BSE effective Hamiltonian; Coulomb / screened Coulomb via ISDF–THC; direct + exchange terms. Diamond, $6 \times 6 \times 6$ k -mesh ($N_k = 216$), point group $|P| = 48$, $n_{\text{ao}} = 26/\text{cell}$. Rank $c5 \Rightarrow n_{\text{IP}} = 136$; $c8 \Rightarrow n_{\text{IP}} = 208$. All counts from the analytic cost model `XPRIZE_paper/script/_resource.dispatch.py` at angle precision $b = 32$, `mode="T-opt"` (per-QROAM select–swap λ chosen for minimum Toffoli). Primitives used: P-06 `unitary_synthesis`, P-07 `forward_isometry_synthesis`, P-09 `diagonal_matrix_block_encoding`, P-12 `eigendecomposition_block_encoding` (ledger `XPRIZE/context/primitives.md`).

Operations (per the model, N_k carried as the QROAM address prefix):

1. **Exchange** $\rightarrow$ eigendecomposition BE of W : 216 blocks of $n_{\text{IP}} \times n_{\text{IP}}$.
2. **Direct** $\rightarrow$ diagonal BE of W : $216 \times n_{\text{IP}}^2$.
3. Occupied X : $6 \times$ isometry $n_{\text{IP}} \times 4$ (216 blocks).
4. Virtual X : $6 \times$ isometry $n_{\text{IP}} \times 22$ (216 blocks).

Raw (k-points only, no point-group symmetry)

	$n_{\text{IP}} = 208$ (c8)	$n_{\text{IP}} = 136$ (c5)
Exchange (eigendecomp)	1,024,873	
Direct (diagonal)	41,655	
$6 \times$ occ isometry	213,768	
$6 \times$ virt isometry	1,175,724	
TOTAL	2,456,020	1,737,854

Symmetry-adapted structure (jsun3 THC_general/log_irreps, diamond $6 \times 6 \times 6$ c5)

- W (grid-grid): **57** whole-space-group irrep blocks (f, μ) ; $\sum m_i n_i = 29,376$; $\sum n_i = 1768$ (copies/eigenvalues); $\sum n_i^2 = 85,864$ (unique W data); $n_i \in [1, 80]$.
- Orbits (Q; diamond_symm_orbit_data.json): **730** orbit-blocks, $\sum |o|^2 = 12,912$; histogram $\{1:104, 2:398, 3:40, 4:1\}$.
- X (grid-ao): **49** blocks; $\sum n_{\text{grid}} n_{\text{ao}} = 17,006$; $n_{\text{grid}} \in [6, 80]$, $n_{\text{ao}} \in [1, 18]$.

Symmetry-adapted construction

Replace X, W by their irrep-block structure; add **6** Q 's (generalized-Fourier / symmetry adaptation, reused across grid legs) + register-recording (family/coset canonicalization). In the symmetry basis both W 's are diagonalized, *but the eigenvector unitary must still be built*:

$$W = U D U^\dagger, \quad U = \bigoplus_i \mathbb{K}_{m_i} \otimes u_i \text{ (copy-space, blocks } n_i).$$

- **Exchange** $W = 2U + D$: U = block `unitary_synthesis` over copy blocks $\{n_i\}$ (group by size, floor<4→4); D = diagonal BE over $\sum n_i$ eigenvalues.
- **Direct** W : diagonal BE over $\sum n_i^2$ reduced values.
- **6** Q : each = orbit synthesis (block `unitary_synthesis` over the orbit histogram) + reconstruction $\approx 2770 \cdot \frac{32}{24}$ (stage-B perm/fold-back, n_k -indep).
- X : $6 \times \text{occ} + 6 \times \text{virt}$ isometries, family-reduced (`forward_isometry_synthesis`($n_{\text{IP}}, M, N_{\text{fam}}=16, b$)).

	c5 ($n_{\text{IP}} = 136$, exact)	c8 ($n_{\text{IP}} = 208$, extrap.)
Exchange $W = 2U + D$	153,115 ($U=76,276, D=563$)	278,162 ($U=138,729, D=704$)
Direct W (diagonal)	3,983	6,099
$6 \times Q$	114,780 ($Q=19,130$)	149,238 ($Q=24,873$)
X isometries	342,264	403,572 (exact@208)
TOTAL	614,142	837,071
reduction vs. raw	2.83×	2.93×

c8 extrapolation: scale multiplicities $n_i \times r$ and orbit counts $\times r$, $r = 208/136 = 1.529$ (both consistent with grid dim $\rightarrow 216 \cdot 208 = 44,928$); #irreps (57) and orbit-size shape held fixed; recompute every cost with the primitives. X computed *exactly* at 208. Scaled: $\sum n_i=2705$, $\sum n_i^2=200,735$, $\sum |o|^2=20,856$.

Key findings

- **W collapses, X does not.** W becomes diagonal $\Rightarrow$ data $\rightarrow$ eigenvalues (direct $W \sim 6k$). X stays an isometry: the M -column rotation structure is irreducible, so symmetry only shrinks the row/data dimension. X reduces $\sim 3.4\times$ (from the k -family fold $216 \rightarrow 16$) and is the **dominant** symmetry-adapted term ($\sim 48\%$).
- **Eigenvector unitary U is real cost.** $W = U D U^\dagger$ needs $U, U^\dagger$; $2U$ is the 2nd-largest term ($\sim 33\%$). (An earlier draft wrongly dropped it to just D .)
- **6 Q 's are cheap** ($\sim 18\%$): the true orbits are mostly tiny (104 size-1, 398 size-2).
- **Block isometry $\nless$ dense-per-family.** Summing the 49 grid-ao blocks independently (425k c5 / 503k c8) *loses amortization* and beats neither; the dense-16-family form is used. A proper shared-column-loop block synthesis (P-15/16, loop to $\max_i n_{\text{band}}$) could trim X to ~ 250 – $350k$ (total ~ 740 – $780k$), not computed here.

Reproduction

```
cd XPRIZE_paper/script; import _resource_dispatch as D; b=32, mode="T-opt", N_k=216, N_fam=16.
Raw: D.eigendecomposition_block_encoding(n_IP, 216, b), D.diagonal_matrix_block_encoding(n_IP, 216, b),
```

D.forward_isometry_synthesis($n_{\text{IP}}, M, 216, b$). Symm: $U = \sum \text{unitary_synthesis}(n_i \text{ grouped by size, count, } b)$; D/direct = diagonal BE via `unranged_qroam_compute_cost+(b-2)+uncompute`, λ T-opt, over table $\sum n_i$ resp. $\sum n_i^2$; $Q_{\text{orbit}} = \sum \text{unitary_synthesis}(\text{orbit sizes grouped})$; $X = \text{forward_isometry_synthesis}(n_{\text{IP}}, M, 1)$. Structure: `log_irreps/data_ov_diamond_6x6x6_c5_grid-grid.txt` (n_i), `..._grid-ao.txt` (X blocks), `diamond_symm_orbi` (orbits).

Caveats: c5 exact, c8 extrapolated; occ/virt split of n_{ao} taken 4:22 proportional; X uses dense-per-family (not the shared-loop block form); reconstruction is a scaled estimate; the whole layer is the analytic model, not an executed circuit.